

Fine-Grained Visual Contrast-Driven Preference Optimization for Mitigating Hallucinations in LVLMs

Anonymous ACL submission

Abstract

Large Vision-Language Models (LVLMs) have shown significant progress on multimodal tasks, yet hallucinations remain a critical limitation to their performance. Traditional Direct Preference Optimization (DPO) can partially mitigate hallucinations, but it overly relies on linguistic priors and neglects visual information. Recent studies often adopt de-visualization or coarse edits to align the visual modality, assuming that "more visual contrast pairs are always beneficial." However, we find that simply adding contrast without considering fine-grained visual cues can actually undermine the gains from text-only preference optimization. To improve visual alignment while preserving language modeling capabilities, we propose Fine-Grained Visual Contrast-Driven Preference Optimization (FG-VPO). An explicit visual perturbation constraint focuses optimization gradients on visual representations and cross-modal alignment. Experimental results on five multimodal benchmarks demonstrate that FG-VPO significantly reduces hallucination rates in large vision-language models while maintaining stable overall performance, outperforming existing state-of-the-art multimodal DPO-based methods across four of these five benchmarks. Our code and data are publicly available at <https://github.com/anonymous260105/FG-VPO>.

1 Introduction

Recently, visual Large Language Models (LVLMs) have made significant progress in tasks such as image captioning (Li et al., 2023a; Chen et al., 2024; Li et al., 2024) and Visual Question Answering (VQA) (Li et al., 2023b; Wu et al., 2024). These models integrate capabilities from the visual modality into the language modality, enabling a single model to process both image and text data and generate natural language responses. However, extensive research and practice indicate that LVLMs

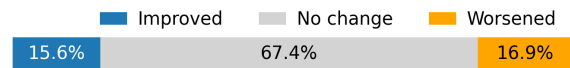


Figure 1: Sample-level CHAIR changes under rejection-conditioned image contrast

remain prone to generating responses inconsistent with image content in real-world complex scenarios—a phenomenon known as multimodal hallucination (Zhong et al., 2024). This issue significantly undermines model reliability. Direct Preference Optimization (DPO) (Rafailov et al., 2023) facilitates preference learning by constructing "preferred–rejected" response pairs without an explicit reward model. This technique has proven effective in large language models and has been naturally extended to LVLMs (Zhou et al.; Fu et al., 2025; Zhang et al., 2024). However, existing multimodal DPO methods generally suffer from the following design limitations: First, training overly relies on textual priors, often with preference supervision dominated by text, while neglecting visual evidence. Second, even when visual information is considered, methods often merely use image removal, blurred images, or coarse-grained object cropping, failing to address fine-grained errors at the attribute and relation levels. Third, there is a widespread assumption that "incorporating visual contrast is inherently beneficial," lacking systematic analysis of the interaction mechanisms between visual contrast design and textual preference objectives (Cui et al., 2023; Wang et al., 2024).

Our research reveals that, under the same textual preference objective, straightforward constructed image–text mixed contrasts do not necessarily yield stable gains and can even counteract the alignment effects achieved by text-only DPO. As shown in fig.1, compared with the text-only DPO baseline,

rejection-conditioned image contrast reduces hallucinations in only 15.6% of samples while worsening them in 16.9%. Although this configuration brings some improvements, counterbalanced by the worsening effects, resulting in an overall slight decline. This phenomenon indicates that "how to construct effective visual contrasts" is itself a core problem that has not been sufficiently explored.

To enhance visual alignment without sacrificing language modeling, we analyze different forms of visual contrast within DPO from an optimization perspective. We note that straightforward image-text mixed contrasts tend to dilute textual gradients and struggle to provide stable and targeted supervision for the visual modality. To improve gradient consistency between text and vision, we propose FG-VPO, a fine-grained visual contrast-driven preference optimization framework for mitigating multimodal hallucinations. FG-VPO structurally organizes multimodal hallucinations into three error types—object, attribute, and relation—and uses this structure to construct fine-grained visual contrast pairs. Furthermore, it combines a diffusion transformer (Esser et al., 2024) to construct visual contrast samples aligned with hallucination patterns and introduces an explicit visual perturbation constraint based on hallucination-aligned visual contrasts within the DPO framework. This encourages the model to rely more on image evidence and enhance cross-modal consistency while retaining its language modeling capability.

The main contributions of this paper are: (1) From optimization and sample-level perspectives, we systematically analyze the interaction between visual contrast and textual preference objectives, revealing that “more visual contrast is not always better,” as simple image-text mixed contrasts may even counteract the gains from textual preferences. (2) We propose fine-grained characterization of visual information at the object, attribute, and relation levels, and construct visual contrast samples aligned with hallucination patterns using diffusion models. (3) We propose the FG-VPO framework, which jointly leverages textual preferences and visual contrast signals within DPO to explicitly enhance visual representations and cross-modal alignment. (4) We validate FG-VPO on five multimodal benchmarks, demonstrating that it significantly reduces the incidence of LVLM hallucinations and stably outperforms representative DPO baselines in visual faithfulness and overall robustness.

2 Related Work

In Large Vision-Language Models (LVLM), preference-based optimization methods have become a leading direction in mitigating multimodal hallucination in LVLMs. Early work primarily extends reinforcement learning from human feedback (RLHF) and direct preference optimization (DPO) from text-only large language models to multimodal settings. For example, RLHF (Sun et al., 2024) achieves vision-language alignment by training reward models that penalize image-text inconsistencies, thereby reducing visual hallucinations. Within DPO-based research, one line of methods constructs preference pairs solely in the textual response space by contrasting faithful and hallucinated outputs. HA-DPO (Zhao et al., 2023) explicitly introduces hallucination-aware terms into the DPO objective and optimizes with hallucination-annotated preference data. POVID (Zhou et al.) augments textual hallucination data with negative samples induced by image perturbations, but the alignment signal still relies on text-side preferences. AMP (Zhang et al., 2024) builds multi-level preference supervision from multiple source models and multi-round evaluations, encouraging the model to follow richer ranking signals. Another line of work further injects perturbations or contrastive cues on the image side. MDPO (Wang et al., 2024) generates negative visual samples via cropping and masking under fixed textual conditions, while V-DPO (Xie et al., 2024) mainly performs object-level contrastive optimization using image-present versus image-absent settings. Overall, these studies demonstrate that preference-optimization-based approaches are effective in suppressing hallucinations in LVLMs. However, existing methods are either text-only or rely on de-visualization and coarse edits. Accordingly, we analyze how visual contrasts interact with textual preferences at both objective and sample levels, and propose a fine-grained visual contrast scheme within DPO for more stable and targeted supervision.

3 Preliminaries

3.1 Large Vision-Language Models

Large Vision-Language Models (LVLM) integrate computer vision and natural language processing technologies to achieve joint understanding and generation of visual and textual modalities. Formally, let the input image be m , the textual instruction or question be x , and the model-generated re-

sponse be y . The LVLM first encodes m into visual embeddings, aligns them with the textual embeddings of x via the adapter, and finally models the conditional distribution $p(y|m, x)$ autoregressively using a language model, thereby accomplishing multimodal understanding and content generation.

3.2 Direct Preference Optimization

Direct Preference Optimization (DPO) is an efficient method for aligning language models with human preferences. Unlike traditional reinforcement learning-based preference optimization methods (such as PPO)(Schulman et al.), DPO directly optimizes over preference-labeled data, thereby avoiding the complicated training pipeline of reward modeling and online RL.

Text-Conditioned DPO In the text-conditioned setting, DPO considers an input prompt x and an output response y . Given a reward function $r(x, y)$, a trainable policy π_θ , and a reference policy π_{ref} , the DPO objective is defined as:

$$\max_{\pi_\theta} \mathbb{E}_{\mathcal{D}, \pi_\theta} [r(x, y)] - \beta \text{KL}[\pi_\theta(y|x) \parallel \pi_{\text{ref}}(y|x)] \quad (1)$$

where β is a temperature hyperparameter and KL denotes the Kullback–Leibler divergence. When this objective is extended to the whole dataset $\mathcal{D}$, the optimal policy can be written as:

$$\pi^*(y|x) = \frac{1}{Z(x)} \pi_{\text{ref}}(y|x) \exp\left(\frac{1}{\beta} r(x, y)\right) \quad (2)$$

where $Z(x) = \sum_y \pi_{\text{ref}}(y|x) \exp\left(\frac{1}{\beta} r(x, y)\right)$ is the partition function. Given a preference dataset $\mathcal{D}_t = \{(x, y_w, y_l)\}$, where y_w denotes the preferred response and y_l denotes the rejected response, DPO optimizes the following loss:

$$\mathcal{L}_{\text{DPO-t}} = -\mathbb{E}_{\mathcal{D}_t} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right) \right] \quad (3)$$

where σ denotes the sigmoid function. This loss encourages the model to increase the relative likelihood of preferred responses while decreasing that of rejected responses, thereby shaping the model’s behavior to better match human preferences.

Vision–Language–Conditioned DPO In the vision–language–conditioned setting, the input is extended from text-only x to an image–text pair

(m, x) . The preference dataset is then given by $\mathcal{D}_v = \{(m, x, y_w, y_l)\}$, and the conditional policies of the trainable and reference models are written as $\pi_\theta(y|m, x)$ and $\pi_{\text{ref}}(y|m, x)$, respectively. The DPO loss keeps the same form, with (m, x) :

$$\mathcal{L}_{\text{DPO-v}} = -\mathbb{E}_{\mathcal{D}_v} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y_w|m, x)}{\pi_{\text{ref}}(y_w|m, x)} - \beta \log \frac{\pi_\theta(y_l|m, x)}{\pi_{\text{ref}}(y_l|m, x)} \right) \right] \quad (4)$$

Note that the log-probability $\pi_\theta(y|m, x)$ implicitly captures the joint influence of both visual and textual modalities. Different designs of visual contrast or augmentation will alter how visual information contributes to the above preference difference term, thus indirectly affecting the model update behavior. This is precisely the aspect that our proposed FG-VPO framework focuses on analyzing and improving in the subsequent sections.

4 FG-VPO

In this section, we first systematically analyze the limitations of existing DPO-based hallucination mitigation schemes for large vision–language models. We then construct targeted datasets by leveraging diffusion models under a fine-grained decomposition of visual factors, in order to more precisely characterize different types of hallucination patterns. Building on these components, we propose the FG-VPO framework (fig.2), which employs fine-grained visual contrast–driven preference optimization to achieve more effective suppression of multimodal hallucinations.

4.1 Problem Analysis

As shown in fig.1 of the introduction, straightforwardly constructed image–text mixed contrasts, compared with text-only DPO, do not yield stable gains. This observation suggests that the issue is not whether to introduce visual contrast, but whether the resulting visual signals provide optimization forces that are directionally consistent with the textual preference objective. We analyze the underlying cause of this phenomenon from the perspective of the optimization objective.

In the LVLM setting, the DPO loss is given in Eq.(4). The update direction is essentially determined by the internal preference gap Δf_θ .

$$\Delta f_\theta = \beta \log \frac{\pi_\theta(y_w|m, x)}{\pi_{\text{ref}}(y_w|m, x)} - \beta \log \frac{\pi_\theta(y_l|m, x)}{\pi_{\text{ref}}(y_l|m, x)} \quad (5)$$

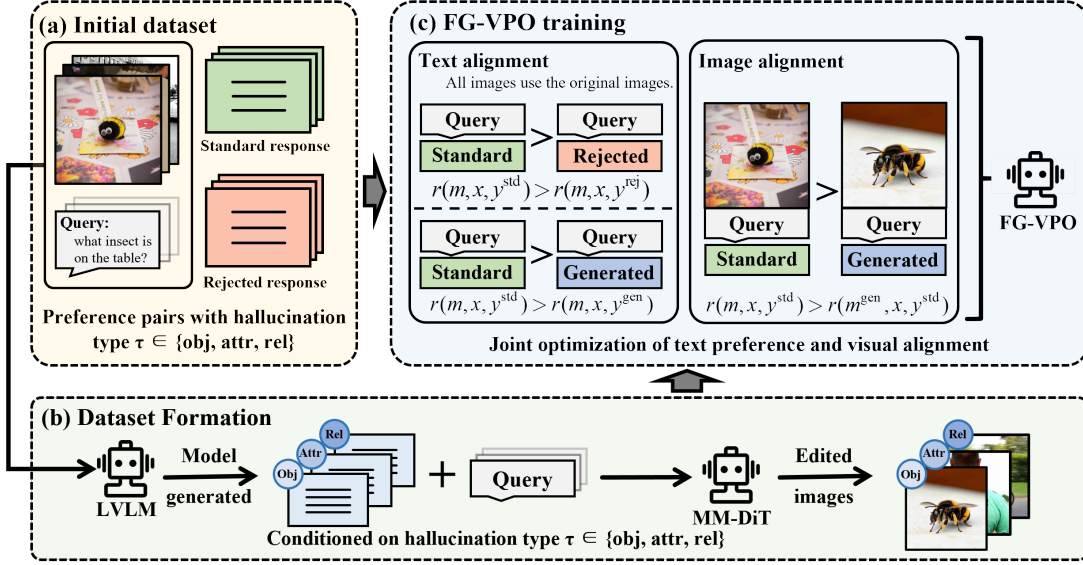


Figure 2: Overall Framework of FG-VPO

When Δf_θ is large, the model clearly prefers y_w and the loss is small; when the model instead prefers y_l or finds the two responses hard to distinguish, the loss increases, thereby pushing the model to adjust its policy.

Since the reference model π_{ref} is fixed, its contribution to Δf_θ is independent of θ and thus does not affect the policy gradient direction. We therefore focus on the policy-dependent component and define $\Delta f_\theta^{\text{pol}} = \log \pi_\theta(y_w|m, x) - \log \pi_\theta(y_l|m, x)$. Here, $\log \pi_\theta(y|m, x)$ reflects the joint influence of language and visual modalities. To analyze how visual contrast designs affect the update direction, we reason at the gradient level and conceptually separate the policy gradient into a language-dominant and a vision-conditioned contribution. From an intuitive perspective, this corresponds to the following first-order interpretive view: $\log \pi_\theta(y|m, x) \approx s_t(x, y) + s_v(m, x, y)$, where $s_t(x, y)$ captures the contribution mainly determined by the input text and response, while $s_v(m, x, y)$ represents the image-dependent cross-modal component. Under this view, the policy-dependent preference gap can be approximated as $\Delta f_\theta^{\text{pol}} \approx \Delta s_t + \Delta s_v$ with $\Delta s_t = s_t(x, y_w) - s_t(x, y_l)$ and $\Delta s_v = s_v(m_w, x, y_w) - s_v(m_l, x, y_l)$.

During training, dominance of Δs_t primarily adjusts linguistic behavior, whereas dominance of Δs_v drives visual alignment. However, in existing multimodal DPO settings, commonly used visual contrast strategies are often weakly aligned with specific hallucination patterns, leading to unstable variations of Δs_v across samples. In some cases,

visual contrast amplifies the textual signal; in others, it interferes with it, causing Δs_v to conflict with Δs_t and resulting in unstable updates.

Therefore, the core challenge of multimodal preference optimization lies in aligning visual and textual preference directions. Based on this analysis, we avoid straightforward image perturbations (such as removal, blurring, or coarse cropping) and instead perform localized, minimally invasive counterfactual edits targeting object-, attribute-, and relation-level hallucinations while keeping other factors unchanged. This approach confines contrastive differences primarily to the dimension of "whether the image supports the standard answer," thereby stabilizing Δs_v and improving its directional consistency with Δs_t , which in turn provides more stable and effective visual supervision for subsequent preference optimization.

4.2 Dataset Formation

As illustrated in fig.3, we first construct an initial dataset $\mathcal{D} = (m, x, y^{\text{std}}, y^{\text{rej}}, \tau)$, where m is the image, x is the textual instruction, y^{std} is the standard response, y^{rej} is a rejected response containing hallucinations, and $\tau \in \{\text{object, attribute, relation}\}$ denotes the hallucination type. This dataset is used for text-only DPO training, where under fixed (m, x) the pair $(y^{\text{std}}, y^{\text{rej}})$ defines a language-level preference relation.

To unify benchmark hallucinations and the model's own hallucinations within a single training domain, we sample from the initial LVLM on (m, x) to obtain a model-generated response y^{gen} .

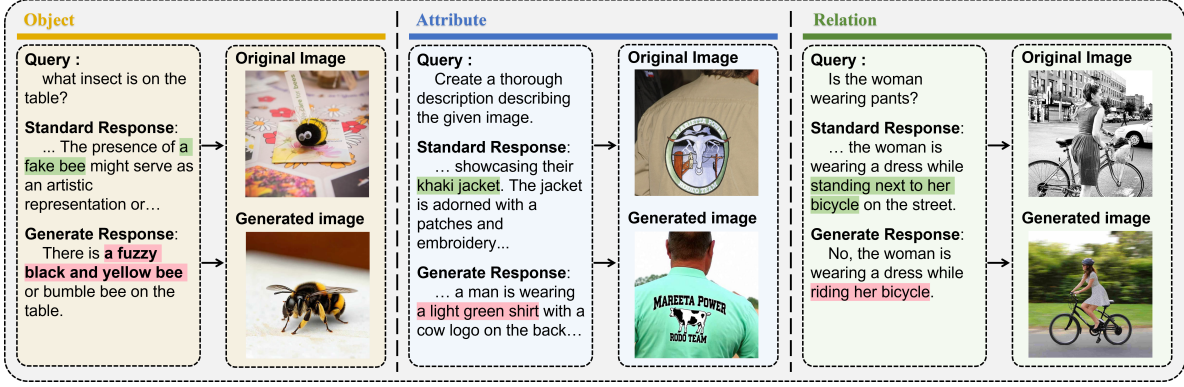


Figure 3: Fine-Grained Hallucination-Aligned Visual Contrasts

Compared with y^{std} , both y^{rej} and y^{gen} may exhibit hallucinations at the object, attribute, or relation level, but they respectively reflect benchmark-curated and model-internal failure modes.

To exploit this information for constructing visual contrasts, we introduce a Diffusion Transformer (MM-DiT)(Esser et al., 2024). Given a query–response pair (x, y) and hallucination type τ , MM-DiT generates a counterfactual image that systematically perturbs the visual evidence along the specified dimension (object, attribute, or relation), while keeping the rest of the scene as unchanged as possible. MM-DiT operates in a shared latent space for images and text, and performs image editing by iteratively denoising latent variables conditioned on (x, y, τ) . Explicit conditioning on the hallucination type τ constrains the edits to be fine-grained and local: object hallucinations correspond to adding/removing/replacing specific entities, attribute hallucinations only modify visual properties such as color or count, and relation hallucinations adjust the geometric layout or interaction patterns between key entities.

Finally, for each sample we construct an image pair (m, m^{gen}) aligned with the standard response y^{std} , and under fixed (x, y^{std}) provide fine-grained, hallucination-aligned visual contrast signals, which lay the foundation for FG-VPO training.

4.3 FG-VPO Framework

We propose FG-VPO (Fine-Grained Visual Preference Optimization), built upon the dataset $\mathcal{D} = (m, m^{\text{gen}}, x, y^{\text{std}}, y^{\text{gen}}, y^{\text{rej}}, \tau)$, with the overall pipeline illustrated in fig.2. The core idea of FG-VPO is to refine the text generation process by aligning both the text and generated responses. While maintaining the model’s alignment with textual preferences, FG-VPO introduces fine-grained

visual contrast signals aligned with hallucination types. These signals encourage the LVLMM to rely more on image evidence during optimization, thus mitigating hallucinations in multimodal settings. To achieve this, we extend the standard DPO framework with a unified training objective that explicitly integrates both textual and visual preferences.

Text Alignment In the textual setting, we retain the conventional preference pairs constructed from $(y^{\text{std}}, y^{\text{rej}})$. Given fixed inputs (m, x) , we define textual preference by contrasting the standard answer with the hallucinatory answer. The corresponding DPO loss is:

$$\mathcal{L}_{\text{text}} = -\mathbb{E}_{\mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y^{\text{std}}|m, x)}{\pi_{\text{ref}}(y^{\text{std}}|m, x)} - \beta \log \frac{\pi_{\theta}(y^{\text{rej}}|m, x)}{\pi_{\text{ref}}(y^{\text{rej}}|m, x)} \right) \right] \quad (6)$$

where π_{θ} denotes the policy model, π_{ref} denotes the reference model, and β is a temperature coefficient. This objective ensures that the model consistently prefers the hallucination-free standard response y^{std} .

Generated Text Alignment We further incorporate preference pairs constructed from the model’s own generated responses y^{gen} to better calibrate its generation behavior. The DPO loss is:

$$\mathcal{L}_{\text{gen}} = -\mathbb{E}_{\mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y^{\text{std}}|m, x)}{\pi_{\text{ref}}(y^{\text{std}}|m, x)} - \beta \log \frac{\pi_{\theta}(y^{\text{gen}}|m, x)}{\pi_{\text{ref}}(y^{\text{gen}}|m, x)} \right) \right] \quad (7)$$

This term further aligns the model’s generative policy during preference optimization toward the distribution of hallucination-free responses.

Visual Alignment To explicitly strengthen the model’s reliance on visual information, we construct image pairs (m, m^{gen}) : m is the original image consistent with the standard response, while m^{gen} incorporates fine-grained perturbations corresponding to hallucination type $\tau \in \{\text{object, attribute, relation}\}$, creating a visual conflict with y^{std} . Given fixed (x, y^{std}) , we treat (m, x, y^{std}) as the preferred sample and $(m^{\text{gen}}, x, y^{\text{std}})$ as the dispreferred sample, yielding the visual DPO loss:

$$\mathcal{L}_{\text{vis}} = -\mathbb{E}_{\mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y^{\text{std}}|m, x)}{\pi_{\text{ref}}(y^{\text{std}}|m, x)} - \beta \log \frac{\pi_{\theta}(y^{\text{std}}|m^{\text{gen}}, x)}{\pi_{\text{ref}}(y^{\text{std}}|m^{\text{gen}}, x)} \right) \right] \quad (8)$$

Unlike Eq.(4), Eq.(8) forms contrast only in the visual modality. This explicitly amplifies the visual preference gap $\Delta s_{\text{vis}}(y^{\text{std}}) = s_{\text{vis}}(m, x, y^{\text{std}}) - s_{\text{vis}}(m^{\text{gen}}, x, y^{\text{std}})$ through gradient updates, allowing it to judge whether an image supports a given answer under identical textual conditions.

Joint Training Finally, the overall FG-VPO objective is formulated as:

$$\mathcal{L}_{\text{FG-VPO}} = \mathcal{L}_{\text{text}} + \lambda_g \mathcal{L}_{\text{gen}} + \lambda_v \mathcal{L}_{\text{vis}} \quad (9)$$

where λ_g and λ_v control the relative contributions of generated-text alignment and visual alignment. During training, we jointly optimize textual and visual preferences, enabling the model to maintain consistent language preference while continuously enlarging the visual preference gap across object, attribute, and relation dimensions. This results in an LVLM with stronger cross-modal consistency and substantially reduced hallucination.

5 Experiments

5.1 Experimental Setup

Models and Datasets We construct our experiments on a subset of the RLAIIF-V dataset (Yu et al., 2024). Specifically, we randomly sample 4.8K instances as the base corpus. Each instance consists of a question, an original image, a human-verified standard response, and a poor response annotated as containing hallucinations. To construct hallucination-aligned visual contrasts, we employ a multi-modal diffusion transformer as the image editing model. This model features a dual-stream design, enabling joint processing of visual and textual inputs for generating counterfactual images.

We utilize LLaVA-v1.5-7B (Liu et al., 2024a) as the base large vision-language model, which is built upon the Vicuna-7B language backbone and integrates a CLIP visual encoder (Radford et al., 2021).

Evaluation Metrics We systematically evaluate model capabilities and hallucination behaviors across five benchmarks: POPE (Li et al., 2023c), AMBER (Wang et al., 2023), TextVQA (Singh et al., 2019), MMBench (Liu et al., 2024b), and CHAIR (Rohrbach et al., 2018). POPE is an object hallucination-oriented benchmark, assessing object existence hallucinations through binary ("yes" or "no") queries about image-question pairs. AMBER is a fine-grained object-annotation-based hallucination benchmark, covering different error types such as objects, attributes, and relations. TextVQA focuses on text-centric visual question answering, requiring models to read and comprehend scene text within images to answer questions. MMBench is a systematically designed comprehensive multiple-choice benchmark covering diverse capabilities including object recognition, text recognition, relational reasoning, and commonsense understanding. MMBench evaluation is performed on standard model outputs from 500 samples.

Baselines We compare FG-VPO with representative DPO-based hallucination mitigation methods, including HA-DPO (Zhao et al., 2023), POVID (Zhou et al.), AMP (Zhang et al., 2024), MDPO (Wang et al., 2024), and V-DPO (Xie et al., 2024). Descriptions are provided in Section 2.

Implementation Details We use LLaVA-v1.5-7B as the base model and train FG-VPO in a distributed manner on two A40 GPUs. During training, the batch size is set to 128, with an effective batch size of 32 per optimization step. We employ LoRA (Hu et al., 2022) fine-tuning with a rank of 256 and a scaling factor $\alpha = 512$. The model is trained for 4 epochs with an initial learning rate of 1×10^{-6} , following a cosine annealing learning rate schedule, and a temperature coefficient of 0.2.

5.2 Overall Results

This section compares FG-VPO with DPO-based hallucination mitigation methods across multiple multimodal benchmarks. The overall results are presented in tab.1, including the baseline model LLaVA-v1.5-7B, standard DPO, as well as HA-DPO, POVID, AMP, MDPO, V-DPO, and our proposed FG-VPO.

Methods	Hallucination						Comprehensive		
	POPE		AMBER			CHAIR		TextVQA	MMBench
	↑	Chair ↓	Cover ↑	HalRate ↓	Cog ↓	Chair _s ↓	Chair _i ↓	↑	↑
LLaVA-v1.5-7B	87.562	7.6	51.8	35.5	4.3	44.0	11.8	57.95	64.30
DPO	<u>87.566</u>	7.4	51.2	32.9	3.8	45.8	12.2	57.91	73.63
HA-DPO	87.348	7.3	52.1	34.1	3.9	43.0	11.5	57.60	<u>73.66</u>
POVID	87.479	9.6	52.4	50.9	7.0	50.4	12.5	58.15	64.90
AMP	85.300	12.7	<u>52.8</u>	56.3	5.3	<u>25.0</u>	<u>10.0</u>	48.23	66.42
MDPO	—	<u>4.4</u>	52.4	<u>24.5</u>	<u>2.4</u>	—	—	—	—
V-DPO	—	5.6	49.7	27.3	2.7	—	—	—	—
FG-VPO (our)	87.568	3.6	52.9	20.2	1.5	21.0	5.7	57.84	73.86

Table 1: Comparison of DPO-based methods on hallucination and comprehensive evaluation for large vision-language models. For baselines with publicly available checkpoints, we re-evaluate them under a unified setting. For MDPO and V-DPO, which do not release checkpoints, we directly report the numbers from the original papers, and results that cannot be obtained are marked with “—”. The best and second-best results are indicated with **bold** and underline, respectively.

From the results on hallucination-related benchmarks including POPE, AMBER, and CHAIR, different preference-optimization methods exhibit clear performance divergence across benchmarks compared to the original LLaVA-v1.5-7B. Specifically, standard DPO achieves modest gains on AMBER but shows slight degradation on CHAIR; HA-DPO demonstrates more balanced performance overall, POVID performs weakly, while AMP excels on CHAIR but performs poorly on AMBER. These results suggest that relying solely on text-level preference construction or training signals is often insufficient to yield consistent cross-benchmark improvements. In contrast, MDPO and V-DPO introduce additional supervision through image-level constructions, but their evaluations are currently limited to the AMBER benchmark, where they show improvements over the baseline with limited overall gains. By comparison, our FG-VPO achieves consistent and significant improvements across POPE, AMBER, and CHAIR, demonstrating more stable and robust cross-benchmark hallucination mitigation capability.

Regarding comprehensive multimodal capabilities (TextVQA and MMBench), differences among methods are relatively moderate overall. On TextVQA, apart from AMP which performs notably worse, other models achieve roughly comparable scores, suggesting that moderate preference alignment does not significantly alter model capabilities in text recognition-centric scenarios. On MMBench, which covers multiple capability dimensions such as object recognition, reasoning, and commonsense, FG-VPO outperforms other DPO-based methods in comprehensive performance, ranking highest overall, indicating its

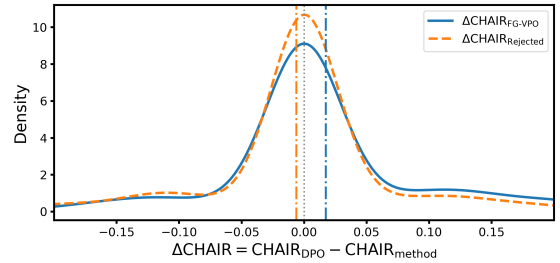


Figure 4: Effect of rejection-conditioned image contrast

sustained competitiveness in broad visual-language understanding and reasoning tasks.

In summary, as shown in tab.1, simply adding or coarsely designing visual contrast within the DPO framework is insufficient to guarantee stable gains across benchmarks. In contrast, the fine-grained, structured visual contrast based on hallucination types enables FG-VPO to achieve optimal or near-optimal overall performance on most evaluation metrics while balancing "hallucination mitigation" and "general multimodal capability." Qualitative examples are provided in Appendix A.

5.3 Sample-level Stability Analysis

To characterize the impact of visual contrast design on model behavior at a finer granularity, we compare the stability of FG-VPO and rejection-conditioned image contrast relative to the text-preference baseline from a sample-level perspective. For each sample, we calculate the relative change in CHAIR: $\Delta\text{CHAIR} = \text{CHAIR}_{\text{DPO}} - \text{CHAIR}_{\text{method}}$. A value of $\Delta\text{CHAIR} > 0$ indicates reduced hallucination in the sample compared to the baseline (improved), $\Delta\text{CHAIR} < 0$ indicates increased hallucination (worsened), and $\Delta\text{CHAIR} = 0$ indicates no change (un-

Methods	AMBER			
	Chair ↓	Cover ↑	HalRate ↓	Cog ↓
LLaVA-v1.5-7B	7.6	51.8	35.5	4.3
Text-Only	5.5	51.3	27.7	2.9
FG-VPO (our)	3.6	52.9	20.2	1.5

Table 2: Ablation Study on Hallucination Mitigation (AMBER)

Methods	CHAIR	
	Chair _s ↓	Chair _i ↓
LLaVA-v1.5-7B	44.0	11.8
Text-Only	32.2	8.8
FG-VPO (our)	21.0	5.7

Table 3: Ablation Study on Hallucination Mitigation (CHAIR)

changed). Fig.4 illustrates the distribution (density) of ΔCHAIR for FG-VPO and rejection-conditioned image contrast. Here, rejection-conditioned image contrast generates contrast images guided by rejection responses and constructs visual preference pairs while keeping the textual input unchanged. The distributions of both methods are centered near zero, indicating that most samples exhibit relatively limited changes. However, the distribution of FG-VPO shows a more distinct positive shift: its mean lies to the right of zero, and its positive tail is longer, meaning that hallucination reduction is observed in more samples while the proportion of samples with negative changes is comparatively smaller. In contrast, the distribution of rejection-conditioned image contrast is more symmetric around zero, with mass in the positive and negative regions being closer, suggesting that it is more likely to simultaneously produce both improvements and deteriorations at the sample level, thereby making its overall gains more reliant on mutual cancellation across different samples.

In summary, under comparable training budgets, FG-VPO—with its fine-grained and explicit alignment to hallucination types—more reliably converts visual contrast signals into sample-level hallucination reduction, rather than merely reflecting a fragile net gain in aggregate metrics.

5.4 Ablation Study

To validate the contribution of different preference signals in FG-VPO, we conduct an ablation study under a unified training and evaluation setup, comparing three configurations: the LLaVA-v1.5-7B baseline; text-only preference alignment, which trains under the DPO framework using two

types of response preference pairs— $(y^{\text{std}}, y^{\text{rej}})$ and $(y^{\text{std}}, y^{\text{gen}})$; and FG-VPO, which jointly optimizes with fine-grained, hallucination-type-aligned visual preferences in addition to the text preferences. We evaluate the above models on the AMBER and CHAIR benchmarks, with results presented in tab.2 and tab.3. Introducing text-only preferences alone yields stable hallucination reduction on both benchmarks compared to the baseline. Building upon this, FG-VPO, which further incorporates visual preferences, achieves additional improvements across all metrics of AMBER and CHAIR, with a consistent upward trend observed across different metrics. These results indicate that text preference alignment provides an effective foundation for hallucination mitigation, while fine-grained visual preferences aligned with hallucination patterns offer complementary supervision, thereby enabling more robust multimodal hallucination suppression.

6 Conclusion

In summary, this paper systematically analyzes the interaction between visual contrast and text preferences at both the optimization objective and sample behavior levels, noting that blindly increasing visual contrast does not necessarily lead to better alignment—simple hybrid designs may even counteract the benefits brought by text preferences. On this basis, we model visual information in a fine-grained manner across three dimensions—objects, attributes, and relations—and leverage a diffusion model to construct counterfactual image pairs explicitly aligned with hallucination patterns. We accordingly propose the FG-VPO framework, which integrates text preference signals with fine-grained visual contrast signals into the DPO optimization process, effectively enhancing visual representation and cross-modal alignment capabilities. Experimental results demonstrate that FG-VPO consistently reduces hallucination rates in LVLMS across five multimodal hallucination benchmarks while maintaining strong general multimodal understanding capabilities.

7 Limitations

Although FG-VPO achieves consistent improvements on multiple benchmarks, several limitations remain. First, the method relies on hallucination annotations at the object, attribute, and relation levels, as well as a fixed diffusion transformer to construct counterfactual images, which imposes

requirements on annotation quality and generative model capability and may limit its applicability in scenarios lacking fine-grained labels or high-quality editors. Second, our current evaluation mainly relies on automatic metrics and has not yet systematically incorporated large-scale human subjective assessment, so the trade-offs between “hallucination reduction” and informativeness or usability have not been fully characterized. Finally, similar to most existing multimodal preference optimization methods, FG-VPO still primarily depends on internal model scores and soft labels, imposing only indirect constraints on hallucination behavior. How to systematically integrate verifiable visual evidence (e.g., independent detectors or external perception modules) into the training process to enforce stronger and more explicit constraints on factual consistency remains an important direction for future work.

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A Qualitative Example

Fig.5 presents several representative qualitative cases comparing the original LLaVA-1.5-7B, DPO, and our proposed FG-VPO. We observe that the original model exhibits various forms of hallucinations at the object, attribute, and relation levels, such as hallucinating extra objects, fabricating colors and material properties, or making overly speculative inferences about scene relationships. Building on this baseline, DPO alleviates part of these errors but still leaves a non-negligible amount of visually inconsistent descriptions. In contrast, FG-VPO tends to ground its generations more tightly in the visible evidence of the image across different scenarios, substantially suppressing fictitious entities and relations, and producing outputs that are more objective and highly consistent with the visual signal.

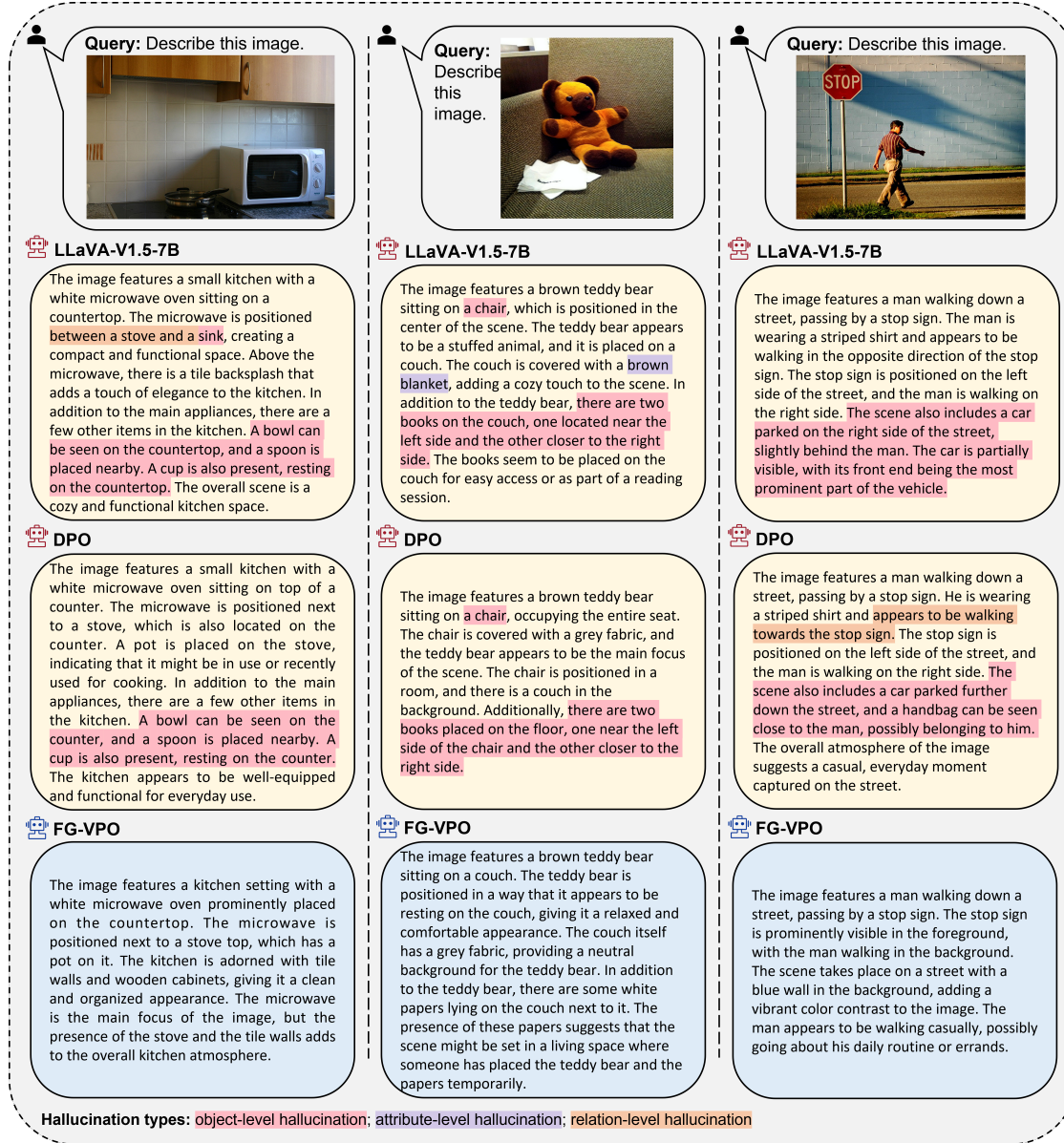


Figure 5: Qualitative Examples of Hallucination Mitigation by FG-VPO